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RESEARCH ARTICLE

Intelligent Anomaly Detection and Localization With Crowd Density Estimation for Improved Public Safety

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ABSTRACT The rapid growth of metropolitan populations and large public gatherings has increased the need for intelligent surveillance systems capable of ensuring public safety. This study proposes a novel framework, Intelligent Anomaly Detection and Localization with Crowd Density Estimation (IADL-CDE), which jointly addresses crowd density classification and anomaly detection in real-time. Leveraging the ShanghaiTech dataset for crowd density estimation and the UCSD Pedestrian Database for anomaly detection, the framework integrates a Robust Maximum Correntropy Kalman Filter (RMCKF) for effective image pre-processing, followed by a Self-Modulating Convolutional Neural Network (SMCNN) that accurately classifies crowd density levels into sparse, medium, and dense categories. Subsequently, a Pixel-Associated Auto Encoder (PAAE) is employed to detect and localize anomalies such as cycling, skateboarding, and unauthorized vehicles in pedestrian environments. Experimental results demonstrate that the proposed IADL-CDE model achieves significant improvements in accuracy, precision, recall, F1-Score and Cohen's Kappa Coefficient compared to the baseline methods. These findings highlight the model's effectiveness and robustness in enhancing public safety through intelligent crowd monitoring and anomaly detection.

INDEX TERMS Anomaly detection, crowd density estimation, localization, pixel-associated auto encoder, robust maximum correntropy Kalman filter, self-modulating convolutional neural network, sustainable cities and communities.

I. INTRODUCTION

Crowd monitoring and management are one of the key aspects of urban development and public safety. In recent years, there has been a growing interest in the expansion of events that attract large numbers of people [1]. The safety concerns surrounding such gatherings have highlighted the need for intelligent crowd monitoring techniques which entails understanding the behaviors and interactions among individuals in large numbers [2], [3], [4], [5]. Accurately detecting and interpreting people's actions as they move and gather is among the most critical security challenges at public events [6], [7], [8], [9]. Congestion and anomaly detection are two highly studied issues in video surveillance [10]. Automated surveillance becomes more important and complex as the activities of human society have expanded [11]. Several activities occurred in both indoor and outdoor settings taking place in low-light conditions with huge shadows [11], [12], [13], [14]. An effective crowd behavior analysis is necessary to spot abnormalities in a crowd and comprehend its behavior by analyzing video data from public surveillance footage [15]. The data gathered is especially relevant for crowd control in public settings depending on certain behaviors [16], [17], [18]. Surveillance equipment technology has evolved dramatically in recent years, improving the quality of geographical data obtained [19]. Event spot analysis is challenging owing to the lengthy viewing distance and low quality of CCTV footage [20], [21], [22]. An anomaly is

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defined as an anomalous movement of a person, a cluster, or a rapid convergence/divergence in a certain region of interest [23], [24], [25]. Also, crowd density level analysis identifies inconsistencies in traffic estimates in surveillance footage [26]. Crowd analysis is classified into various categories, including density, anomalous behavior, crime scene, crowd counting, and event interpretation [27]. To comprehend behaviors, group activity identification and event interpretation require much improved spatial temporal model information. The fundamental approaches of understanding crowd behavior depend on traditional spatial-temporal characteristics like handmade and convolution kernel characteristics [28]. Traditional video surveillance systems rely on real-time monitoring and post-event analysis to detect security threats and anomalies. However, both approaches demand continuous human interaction leading to visual impairment and psychological strain for surveillance personnel. This reliance not only affects efficiency but also raises the possibility of missing key events owing to human limitations. The necessity for fast response to unanticipated occurrences and unusual activities complicates the monitoring work. Dynamic crowd behavior and varying density levels complicate consistent pattern recognition, often leading to missed and false detections. To overcome these issues, using PAAE combined with crowd density estimates allows for accurate detection and localization of anomalies even in crowded environments. This integrated approach improves situational awareness and supports timely interventions to ensure public safety.

The novelty of the proposed research lies in the synergistic combination of PAAE with Crowd Density Estimation for intelligent anomaly detection and precise localization in crowded environment. The model leverages SMCNN to dynamically classify crowd density levels enabling context aware anomaly analysis. PAAE operates to detect and localize typical events by clarifying spatial pixel associations that distinguish normal pedestrian behaviors from anomalies. This dual module architecture enables real-time adaptive safety monitoring with enhanced situational awareness marking significant advancement in public safety surveillance schemes. The main aim of this research is to develop an enhanced technique that assures accurate detection and localization of anomalous activity in crowded environments. It aims to improve public safety surveillance by combining crowd density classification and anomaly detection into a unified framework. This approach improves scene understanding by combining density aware analysis with exact anomaly localization using advanced deep learning technique.

Major contribution of this paper includes,

- The objective of the proposed research is to create an intelligent anomaly identification and localization system by integrating PAAE with crowd density estimation to improve safety monitoring in crowded environments.
- The datasets provide a variety of crowd and anomaly environments that are required for training and evaluating the proposed system. ShanghaiTech provides crowd density estimation, while UCSD provides anomaly detection and localization.
- Utilized RMCKF to effectively resize input images and improving the quality of both crowd density classification and anomaly detection.
- Employed SMCNN to accurately classify crowd density estimation into sparse, medium and dense categories enabling adaptive anomaly detection strategies based on contextual crowd conditions.
- To detect and localize anomalies utilizing PAAE and identify anomalies such as skateboarding, cycling, vehicular intrusions and wheelchair movement ensuring fine-grained anomaly localization at the pixel level.
- The combined framework leverages the synergy between crowd density estimation and anomaly localization to enhance detection reliability for improved public safety in crowded environments.

The remaining paper is configured as follows: Section II provides an in-depth literature review, discussing relevant studies and methodologies. Section III presents the proposed methodology, detailing the techniques and algorithms utilized. Section IV shows the results, including performance evaluations and offers a comprehensive discussion of the findings. Finally, Section V concludes the study by summarizing key insights and suggesting directions for future research.

II. LITERATURE SURVEY

Several research were presented in the literature based on anomaly detection and localization and crowd density estimation in crowded environments utilizing deep learning techniques.

In 2024, Li et al. [29] presented a weakly supervised crowd density estimation technique under two-stage linear feature calibration. To provide a new training technique, the correlation between global and local image attributes was employed to train the network using incremental learning. A parent-child network (PC-Net) that focuses on global and local images was presented. It also includes a linear feature calibration module that trains PC-Net concurrently by applying transfer factors and bias weights to adjust the features extracted from the parent network.

In 2024, Fotia et al. [30] presented an optimization technique for counting people in congested situations based on curriculum learning. It proposed a training technique for anticipating many people present and assuring exact counting accuracy in moderately and densely packed settings. A problem formulation depending on point detection with each point representing an individual head was presented. It uses Curriculum Learning (CL) to recreate progressive learning procedure found in human cognition, starting with simple tasks and progressing to increasingly complex tasks as training progresses.

In 2024, Guo et al. [31] presented a multiscale aggregation network for crowd counting based on a smooth inverse map.

To interchange the usual Gaussian density map with more efficient option, the smooth inverses map (SIM). To aid in model stability, SIM provides a smooth gradient and spatially represents the head location. Then, to improve semantic information across a variety of receptive fields, it employs multiscale aggregation (MA) to adaptively fuse properties across several hierarchies. SIM and MA were used as complementing modules to assist the technique in learning an appropriate density map.

In 2024, Niaz et al. [32] presented spatially aware fusion in a 3D convolutional autoencoder to detect video anomalies. The unsupervised end-to-end design uses geographical and temporal parameters to recognize anomalies in surveillance data. Using a 3D convolutional autoencoder with an encoder-decoder structure, the technique learns spatiotemporal representations. It uses latent space to reconstruct the input. Skip links between encoder and decoder blocks let information flow across multiple scales of feature representation, hence improving the reconstruction process and overall performance.

In 2024, Ul Amin et al. [33] presented video anomaly identification depending on efficient spatiotemporal feature fusion with three dimensional Convolutions and Long Short-Term Memory (LSTM) modules. A new method for capturing spatiotemporal features in video data that combines 3D convolutions and LSTM modules was presented. Unlike typical convolutional neural networks, the method used depth-wise feature stacking to reduce processing costs while enhancing architecture. It uses micro autoencoder blocks for down sampling, reduces the computational complexity of ConvLSTM2D layers, and leverages frequent feature concatenation blocks for up-sampling to maintain temporal information.

In 2025, Akpulat and Ekinci [34] presented anomaly detection in crowd scenes through cross trajectories. Finite-time braid entropy (FTBE) theorem was used for extracting feature vectors to identify aberrant events in crowds. Detecting unexpected occurrences in crowd movements was an important study in community safety and management. A trajectory-dependent technique to identify anomalous occurrences was given under FTBE theorem including motion vector distribution. The image was split into equivalent parts, and the response was obtained through categorizing each activity to many zones.

In 2021, Gao et al. [35] have presented domain-adaptive crowd counting (DACC) through higher quality image translation with density reconstruction. It suggests DACC, which has a higher quality image translation including density map reconstruction. Specifically, by separating domain-shared and independent characteristics and creating content-aware consistency loss, the former concentrates on converting synthetic data into realistic visuals, which improves translation quality. Then the pseudo labels were used to retrain the final counter after that. Testing of adaptation on six real-world datasets show that the suggested approach performs better than the most advanced techniques.

In 2024, Zhang et al. [36] have suggested learning long-range relationships for temporal aircraft anomaly detection. It suggested the stratified multiscale representation learning network with automatic data augmentation (SMDA-Net) for long-range time-series modeling. Specifically, it designed a stratified structure to extract multiscale characteristics of time-series, wherein develop an encoder with an efficient self-attention block for ultra-long sequences. Meanwhile, it presented a learning scheme to weigh the contribution of the augmented samples to the loss for automatic data augmentation that improve the generalization ability of the model.

In 2024, Gao et al. [37] presented dynamic proxy domain that uses binary segmentation to generalize the crowd localization. It suggested a Dynamic Proxy Domain (DPD) method to generalize the learner under domain shift. DPD algorithm which is composed by a training paradigm and proxy domain generator to enhance the domain generalization of the confidence-threshold learner.

In 2020, Gao et al. [38] have suggested feature-aware adaptation and density alignment for crowd counting in video surveillance, which adapt the model from synthetic data to specific real-world scenes. It includes structured density map alignment (SDA) and multilevel feature-aware adaptation (MFA). Specifically, MFA enhances the model to extract characteristics from many levels that were domain-invariant. SDA ensures that the network produces fine density maps on the real domain that are distributed reasonably. Table 1 shows a comparison of literature surveys.

Table 1 summarizes the various advanced crowd analysis and anomaly detection methods. Many challenges remain, despite the crowd counting, anomaly detection, and significant progress in traffic analysis using advanced neural networks and diffusion models. The current methods depend a lot on local or global features, but rarely integrate efficiently, limiting accuracy in complex, dense scenes. Weak supervision and unsupervised learning approaches often face inaccurate localization and difficulties in detection. In addition, high computational cost of multiscale and hybrid transformer architecture restricts real-time applicability. Data augmentation sometimes produces unrealistic samples that affect models generalization. Finally, methods of detecting multiple anomalies depend on trajectory data, which miss subtle or cross-zone anomalies. Addressing these gaps by developing the models that effectively fuse the multiscale spatial-temporal features with robust, significantly enhance accuracy and efficiency. This enables the system to detect more reliable crowd analysis and anomaly detection for real-time applications in complex environments.

III. PROPOSED METHODOLOGY

This section discusses the proposed IADL-CDE method focuses on detecting and classifying crowd density while

TABLE 1. Comparative table of Literature Survey.

Author (Year)	Method	Objective	Advantage	Disadvantage
Li, Y.C., et al. (2024)	Weakly supervised crowd density estimation with Parent-Child Network (PC-Net) and two-stage linear feature calibration	Improve density estimation using correlation between global and local features with incremental learning	Leverages both global and local contexts; enhances feature calibration	Weak supervision may limit accuracy in highly complex scenes
Fotia, L., et al. (2024)	Curriculum Learning (CL) for crowd counting	Enable progressive learning from simple to complex crowd counting tasks	Mimics human cognition; improves accuracy in dense crowds	Training may be slow due to staged complexity
Guo, X., et al. (2024)	Multiscale Aggregation (MA) with Smooth Inverse Map (SIM)	Replace Gaussian density maps with smoother alternatives for stability	SIM ensures smooth gradients; MA enhances semantic fusion	Increased complexity may raise computational cost
Niaz, A., et al. (2024)	3D Convolutional Autoencoder with spatially aware fusion	Detect video anomalies using spatiotemporal features	Captures both spatial and temporal patterns; effective reconstruction	Computationally expensive; needs large datasets
Ul Amin, S., et al. (2024)	3D Convolutions + LSTM with micro autoencoder blocks	Capture spatiotemporal features for video anomaly detection	Efficient spatiotemporal fusion; reduced ConvLSTM2D complexity	Depth-wise stacking may still struggle with large-scale datasets
Akpulat, M. & Ekinci, M. (2025)	Cross-trajectory anomaly detection using Finite-Time Braid Entropy (FTBE)	Detect abnormal events in crowd movements via trajectory analysis	Captures global and local crowd dynamics; zone-based activity detection	Relies heavily on accurate trajectory extraction
Gao, J., et al. (2021)	Domain-Adaptive Crowd Counting (DACC) with image translation and density reconstruction	Adapt models across domains using synthetic-to-real translation	Effective domain adaptation; strong performance across datasets	Depends on quality of translated images; pseudo-labeling errors possible
Zhang, D., et al. (2024)	SMDA-Net (Stratified Multiscale Representation) with automatic data augmentation	Temporal anomaly detection in aircraft data	Captures long-range dependencies; automated augmentation improves generalization	Tailored to time-series, less direct for spatial crowd data
Gao, J., et al. (2024)	Dynamic Proxy Domain (DPD) generalization	Enhance crowd localization under domain shift	Reduces generalization error; robust binary segmentation	Requires careful proxy domain generation
Gao, J., et al. (2020)	Feature-aware Adaptation + Structured Density Alignment	Adapt synthetic-to-real models for video surveillance crowd counting	Extracts domain-invariant features; ensures realistic density maps	Complex training; domain alignment requires fine-tuning

simultaneously identifying and localizing anomalies in crowded environments to improve public safety. This process involves three key steps: Image Acquisition, Pre-processing, Classification and Localization. Initially, images are collected from the ShanghaiTech dataset for crowd density estimation and from the UCSD Pedestrian Database for anomaly detection. The images are resized and normalized using RMCKF in the pre-processing phase. The processed imageries from ShanghaiTech dataset are input into SMCNN to classify crowd density into sparse, medium and dense categories. The images from the UCSD Pedestrian Database are analyzed using PAAE, which identifies anomalies such as cycling, skateboarding, vehicles and wheelchairs based on pixel-level reconstruction errors. This integrated framework ensures both accurate crowd density classification and precise anomaly localization enhancing the overall effectiveness of crowd surveillance systems. In the proposed method, the adaptive graph refinement process dynamically updates the node relationships through an attention- guided similarity mechanism that considers the spatial-temporal feature similarity, temporary co-occurrence pattern, and density-inconvenience to emphasize important connections to suppress irrelevant ones. In each refinement phase, the edge weight is reorganized using a learnable transformation of these combined scores and is normalized through a softmax operation, allowing the graph structure to adapt to the real-time to develop crowds and anomaly patterns. This dynamic update improves message proliferation in relevant nodes, increasing the accuracy of detection. Fig. 1 represents a block diagram of proposed IADL-CDE approach. Accordingly, detailed description of all steps is provided below.

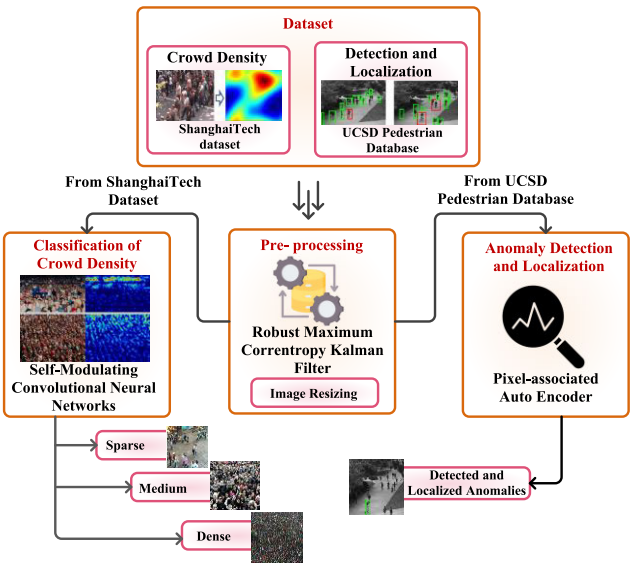


FIGURE 1. Block diagram for proposed of IADL-CDE method.

A. IMAGE ACQUISITION

This section discusses dataset descriptions ShanghaiTech Dataset [39] and UCSD Pedestrian Database [40].

1) SHANGHAITECH DATASET

The ShanghaiTech crowd counting dataset is a larger scale benchmark for crowd counting and density estimation that includes 1,198 images and 330,165 tagged people. It is separated into two parts, Part A contains 482 high density crowd images sourced from the Internet, while Part B comprises 716 low density pictures from Shanghai streets. Each image is tagged with head locations and accompanied by a density map, making it possible to train and evaluate crowd counting models simultaneously. The dataset is widely used in research to build benchmark algorithms capable of handling a variety of crowded environments and densities. A sample image from the ShanghaiTech Dataset and its corresponding density map are shown in Fig. 2, while Fig. 3 presents the class distribution in the training and validation sets.

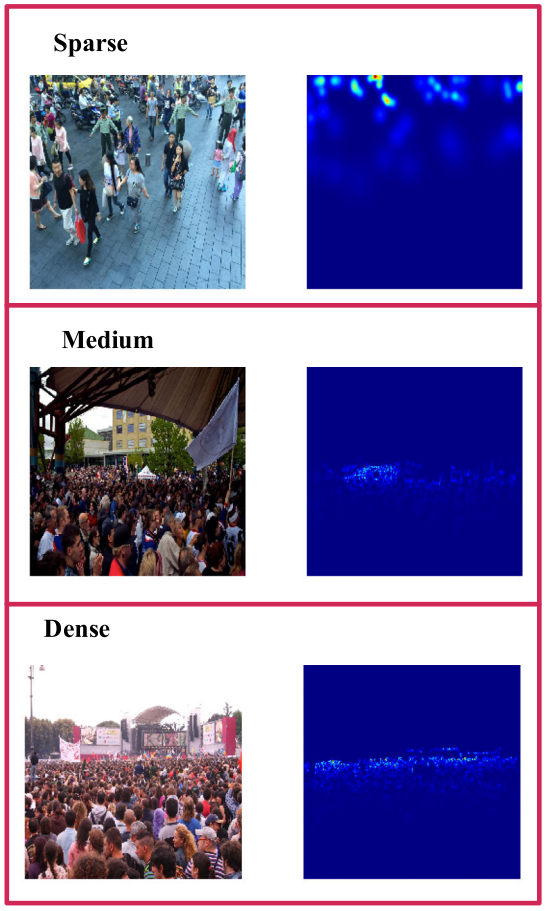


FIGURE 2. Sample images from the ShanghaiTech dataset along with their corresponding density maps.

2) UCSD PEDESTRIAN DATABASE

UCSD Pedestrian dataset is a video dataset for research on pedestrian detection, tracking and crowd analysis. It consists

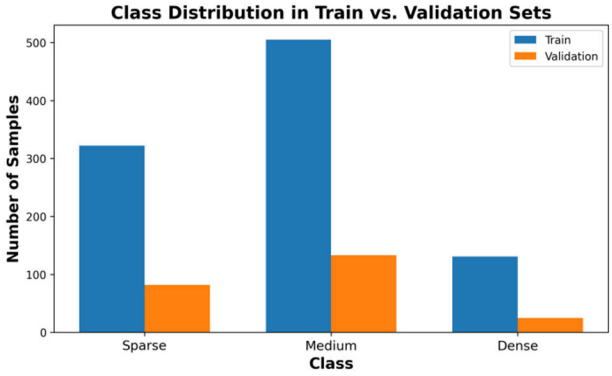


FIGURE 3. Class distribution in train and validation sets.

of video clips shot by a stationary camera positioned over a pedestrian sidewalk at the University of California, San Diego. It includes nearly 2,000 frames with precise annotations, such as pedestrian positions (bounding boxes). It has varying crowd densities, occlusions and perspective distortions, making it perfect for building and testing algorithms for pedestrian detection, tracking and crowd counting in real-world environments.

B. PRE- PROCESSING UTILIZING ROBUST MAXIMUM CORRENTROPY KALMAN FILTER

In this section, RMCKF [41] is discussed to resize the image from input data. RMCKF robustness ensures that the scaling process keeps critical image information while reducing distortion, resulting in higher-quality outputs than older approaches. The preprocessing step uses the RMCKF to increase the image quality by reducing noise. This filter preserves important details by consistently resizing images, ensuring clear and more stable input. Unlike standard filtering methods, RMCKF effectively preserves important structural details and enhances signal reliability, which is essential for accurate crowd density estimation and anomaly detection. The input data of RMCKF is given in (1).

$$E_k = X_k Y_k + f_k \tag{1}$$

Here, E represents the input image; k denotes the raw image data from sources; Y indicates the images resolution; f represents the image quality and X indicates the input image process. The image data is analyzed to estimate the level and type of noise present and RMCKF is particularly suited for environments where traditional Gaussian assumptions fail. To estimate the level of image data is calculated in (2).

$$P_L\left(Y \rightleftarrows_k\right) = \frac{1}{L}\sum_{i=1}^{L}\left[\exp\left\{-\sum_{j=1}^{k-1}\frac{\sigma_1 f_{j,i}^2}{2\mu^2} - \frac{\sigma_2 f_{k,i}^2}{2\mu^2}\right\}\right] \tag{2}$$

where, P indicates the image data to assess noise type and intensity; Y represents the detected non-Gaussian noise patterns; L denotes the total length of image data; σ_1 and σ_2 indicates the image for robust filtering by understanding its

distortion characteristics and μ represents the configuring RMCKF to adapt effectively. The correntropy kernel size is also selected to balance sensitivity to outliers and noise suppression. This careful parameter setup enables the filter to effectively handle non-Gaussian noise during the resizing process, and it is given in (3).

$$C_k = \begin{bmatrix} C_{p,k|k-1} & 0 \\ 0 & C_{r,k} \end{bmatrix} \tag{3}$$

Here, C represents the size balances sensitivity to outliers; r indicates the performance during the image resizing process and p denotes the covariance model uncertainty in image data. RMCKF is applied iteratively to each image, and it estimates true pixel values while simultaneously resizing the image to a target resolution, and it is given in (4).

$$f_{k,i} = e_{k,i} - x_{k,i} y_{k,i} \tag{4}$$

Where, e denotes the iteratively to each image; x represents the true pixel values and y indicates corrects deviations for accurate and clear resizing. The final output is a resize image where structural integrity and fine details are preserved better and ensuring high-quality images for further use. To resize the image is given in (5).

$$\hat{Y}_{k|k-1} = G_k Y_{k-1|k-1} \tag{5}$$

Here, $\hat{Y}$ represents the output of the image and G indicates the resized image. By processing, the RMCKF method resized the image in the input data. Fig. 4 and 5 shows pre- processed images from ShanghaiTech dataset and UCSD pedestrian database respectively.



FIGURE 4. Pre- processed images from ShanghaiTech dataset.

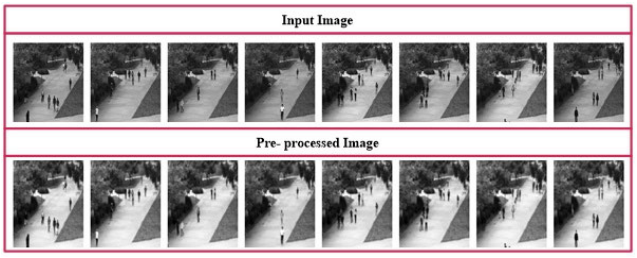


FIGURE 5. Pre- processed Images from UCSD pedestrian database.

Then the pre-processing image is fed to classification of Crowd Density and Anomaly Detection and Localization phase.

C. CLASSIFICATION OF CROWD DENSITY ESTIMATION UTILIZING SELF-MODULATING CONVOLUTIONAL NEURAL NETWORKS

In this section, SMCNN [42] is used to discuss for classifying crowd density as sparse, medium and dense. SMCNN for identifying crowd density relies on adaptive feature modulation, which increases the network sensitivity to changing crowd patterns. The density of the crowd varies from sparse gatherings to highly congested areas requiring a model that is suited for diverse feature distribution. SMCNN was chosen for its self-modulating capacity to dynamically adjust, which enables strong classification at the level of varying density levels. The crowd is classified as sparse, medium and dense to effectively represent different crowd conditions and enable more accurate analysis. These categories help the model distinguish between low, medium and high crowd concentrations, which is necessary to understand the congestion of the crowd and identify potential security risks. By classifying the level of crowd density, the system is better favorable for different scenarios and provides timely alerts for overcrowding or unusual fluctuations. This approach dynamically adjusts convolutional filters based on input features, allowing the technique to focus on key spatial and contextual information unique to varying crowd densities. SMCNN improves accuracy in detecting subtle fluctuations in crowd distribution. It effectively handles variations in scale and viewpoint and reduces the impact of background clutter. Fig. 6 displays the SMCNN architecture for classification of crowd density. It represents a neural network for processing spatial and spectral data, likely for hyperspectral imaging. It starts with separate inputs for spatial and spectral data; each processed through 20 × 20×20 blocks with ReLU activation.

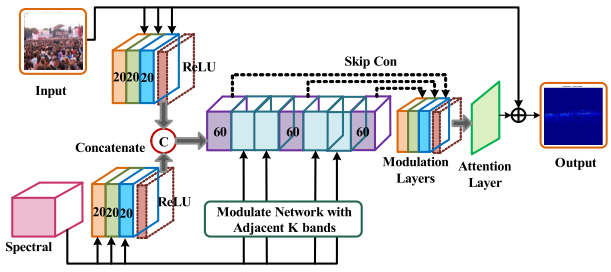


FIGURE 6. Architecture diagram for SMCNN.

These inputs are concatenated into a 60-channel feature map, which then passes through modulation layers incorporating adjacent spectral bands. The output goes through an attention layer to highlight important structures, with a skip connection from the concatenation stage and the result is passed to the output layer. Preprocessed images from the ShanghaiTech dataset are used as input containing crowd density and is given in (6).

$$Y = [Y_1, Y_2, \ldots, Y_i, \ldots, Y_B] \tag{6}$$

Here, Y represents input layer of the classification; Y_i denotes classify the crowd density of ith image and Y_B indicates the full spectrum of crowd densities. Multiple convolutional layers are applied to learn deep spatial representations of crowd patterns. These layers detect crowd density and structural information and is given in (7).

$$Z = [Z_1, Z_2, \ldots, Z_i, \ldots, Z_B] \tag{7}$$

Here, Z represents image classification in modulation layer; Z_i indicates the deep spatial representations of crowd patterns of ith image and Z_B represents the capture crowd density variations. This core module adaptively modulates feature maps using scene-dependent parameters. The network dynamically alters its internal response to different image regions is given in (8).

$$\omega_c = \frac{1}{ab} \sum_p^a \sum_q^b g_{pre}^c (p, q) \tag{8}$$

Here, ω_c represents the standard deviation of classification; ab denotes the self-modulation parameters attained pixel-wisely from attention layers; p and q indicates dynamically adjusts response to different image regions and g_{pre}^c represents the image core module in SMCNN. The aggregated features are processed through fully connected layers that output a final class prediction, labeling the image based on the learned crowd density patterns. Classifying crowd density is given in (9).

$$\phi_c = \sqrt{\frac{1}{ab} \sum_p^a \sum_q^b \left(g_{pre}^c (p, q) - \omega_c \right) + \varepsilon} \tag{9}$$

Here, ϕ_c denotes the classification of output image and ε represents the crowd density patterns in self-modulating layer. Fig. 7 shows crowd density estimated samples by SMCNN model.

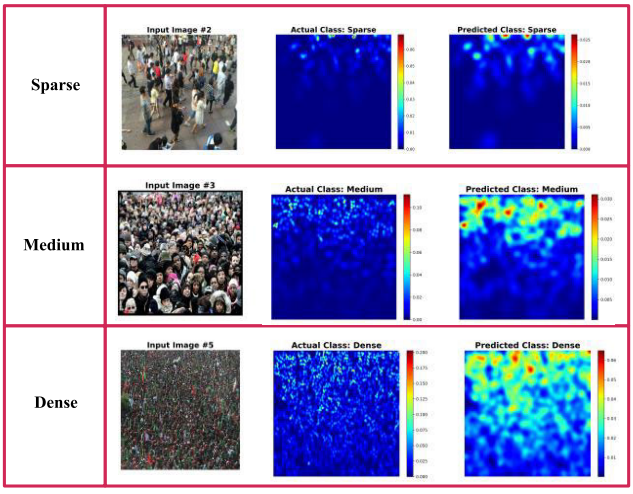


FIGURE 7. Crowd density estimated samples by SMCNN model.

D. ANOMALY DETECTION AND LOCALIZATION UTILIZING PIXEL-ASSOCIATED AUTO ENCODER

In this section, the PAAE [43] is used to discuss anomaly detection and localization. PAAE identifies anomalies such as cycling, skateboarding, vehicles and wheelchairs in pedestrian scenes. The anomalies in pedestrian environments are subtle and often localized such as bicycles, skateboards, or vehicles in restricted areas. PAAE was employed because its pixel-level reconstruction error highlights unusual patterns with high sensitivity. This allows not only to detect but also allows accurate localization of anomalies improves interpretation to the deployment of the real world. PAAE for detecting and localizing anomalies lies in its capacity to correctly recognize irregularities at the pixel level allowing for very accurate and fine-grained localization even in congested environments. This approach uses an autoencoder to learn common pedestrian movement patterns and reproduce typical environments. The reconstruction error increases dramatically at the anomalous pixels, highlighting the abnormalities. PAAE not only detects the presence of anomalies but also precisely identifies exact positions, boosting detection sensitivity and lowering false positives when compared to standard patch-based approach. The Fig. 8 illustrates the architecture of PAAE.

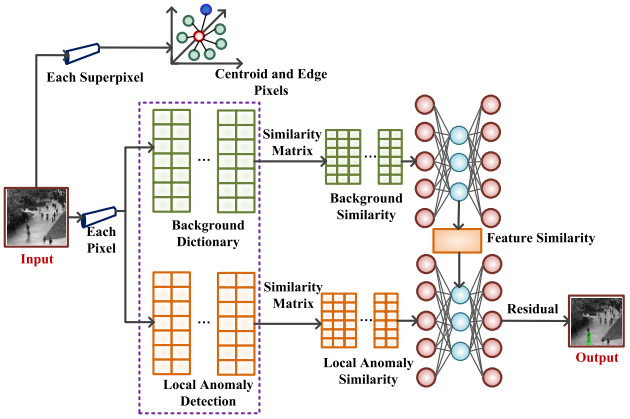


FIGURE 8. Architecture diagram for PAAE.

The PAAE architecture for anomaly detection and localization is displayed on Fig. 8. It starts with an input image from which both individual pixels and super pixels are retrieved. Super pixels are used to create centroid and edge pixels. These attributes are then fed into two separate components as background dictionary represents in green boxes and local anomaly detection module is represented in orange boxes both of which output similarity matrices. These matrices, which represent local anomaly similarity, background similarity, are used as inputs to a multi-layered neural network structure like an autoencoder. This network processes similar features, possibly learning a compressed representation and then recreating it. Residual output is derived from the difference among input and reconstructed output essential

for anomaly identification since it highlights abnormal locations in the final output image. Preprocessed images from the UCSD Pedestrian Database are used as input containing pedestrian scenes is given in (10).

$$\boldsymbol{\alpha}_i^{'}=\frac{\sum_{j=1}^{k}\sum_{k=1}^{K}t_{k=1}^{k}\boldsymbol{\alpha}_j}{\sum_{j=1}^{k}\sum_{k=1}^{K}t_{k=1}^{k}} \tag{10}$$

Here, $\boldsymbol{\alpha}$ represents the input for anomaly detection and localization; $\boldsymbol{i,j}$ and $\boldsymbol{k}$ denotes the input image includes pedestrian scenes and $\boldsymbol{K}$ indicates the anomaly conditions. The encoder processes the input image for extracting meaningful features. It compresses image to lower dimensional latent representation. This representation captures patterns related to normal pedestrian behavior and it is given in (11).

$$\boldsymbol{\beta}_i^{'}=\frac{\sum_{j=1}^{k}\sum_{k=1}^{K}t_{k=1}^{k}\boldsymbol{\beta}_j}{\sum_{j=1}^{k}\sum_{k=1}^{K}t_{k=1}^{k}} \tag{11}$$

Here, $\boldsymbol{\beta}$ denotes the encoder processes each image. Decoder reconstructs the image from latent representation and technique is trained only on normal patterns, it performs accurate reconstruction for regular behavior. This difference in reconstruction helps to identify abnormal events in the scene and it is given in (12).

$$u_j=\arccos\left(\frac{t_j^S d_k}{\sqrt{t_j^S t_j}\sqrt{d_k^S d_k}}\right) \tag{12}$$

Here, $\boldsymbol{\mu}$ represents the decoder reconstructs the image; $\boldsymbol{S}$ denotes trained exclusively on normal behavior patterns and $\boldsymbol{d}$ indicates the identification of deviations from normal behavior. The difference among the input and reconstructed image is computed at the pixel level and areas with high reconstruction error indicate abnormal behavior or unseen patterns. The difference between input and reconstructed image is expressed in (13).

$$f_{shb}(k)=\sqrt{\left(\boldsymbol{\alpha}_{tp}-j_{bg}\right)^2+\left(\boldsymbol{\beta}_{tp}-k_{bg}\right)^2} \tag{13}$$

Here, f_{shb} indicates the difference among input as well as reconstructed image; $\boldsymbol{\alpha}_{tp}$ and $\boldsymbol{\beta}_{tp}$ represents the presence of unseen and unusual patterns; $\boldsymbol{j}_{bg}$ denotes difference calculated at the pixel level and $\boldsymbol{k}_{bg}$ represents the localized image. Anomalies movements are detected where reconstruction error is significantly high. These regions are then localized precisely within the frame using pixel level error maps. The anomaly detection and localization is given in (14).

$$q_n=\frac{1}{1+w^{-zn}} \tag{14}$$

Here, $\boldsymbol{q}_n$ denotes the output of anomaly detection and localization; $\boldsymbol{w}$ indicates the anomalous region's identification and $\boldsymbol{zn}$ represents the unusual movements. Fig. 9 shows detected and localized anomaly by PAAE. Finally, PAAE has successfully detected and localized anomalies.

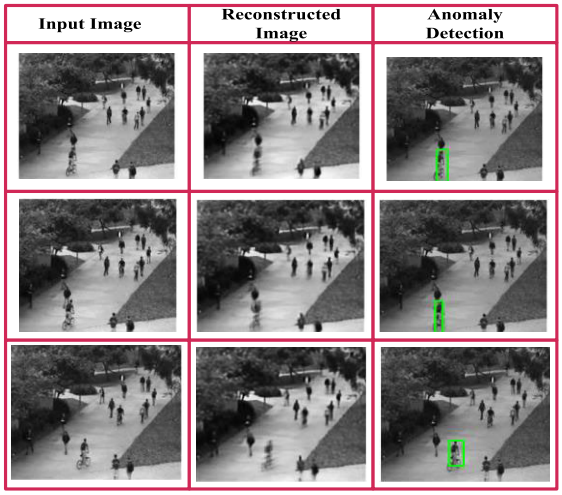


FIGURE 9. Detected and localized anomaly by PAAE.

IV. RESULTS AND DISCUSSION

This section discusses the simulation results of the proposed IADL-CDE method. The model was implemented in Python and experiments run on the NVIDIA RTX 3090 (24 GB) and evaluated using several performance metrics to assess its effectiveness. For ShanghaiTech dataset, the proposed approach was compared with WCDE-TLFC- PC-Net [29], CC-DE-CLT [30] and MAN-SIM-CC [31]. For UCSD Pedestrian Database the proposed approach is compared with SAF-3DCAE-VAD [32], VAD-3DC-LSTM [33] and AD-CS-FCNN [34] approaches.

TABLE 2. Simulation parameter.

Parameter	Value
Training Epochs	50
Batch Size	32
Optimizer	Adam
Learning Rate – SMCNN	0.001
Learning Rate – PAAE	0.0005
Activation Function	ReLU
Kernel Size	5
RMCKF – Correntropy Width	0.8
Adaptive Graph – Neighbors per Node (k)	8
Graph Convolution Orders	3
Weight Initialization	Xavier Uniform
Loss Function – SMCNN	Cross-Entropy
Loss Function – PAAE	MSE Reconstruction Loss

Table 2 displays the simulation parameters of the IADL-CDE model. The network is trained for 50 epochs with a batch size of 32 using Adam optimizer with separate learning rates of 0.001 and 0.0005 for SMCNN and PAAE respectively. The preprocessing step i.e. RMCKF employed the kernel size of 5 and the correntropy width of 0.8, while the adaptive graph retained 8 neighbors per node with two graph convolution orders. Xavier Uniform initialization, ReLU Activation and appropriate loss functions were implemented.

Table 3 presented cross-dataset evaluation results, highlighting the generalization ability of the proposed model.

TABLE 3. Cross-dataset evaluation results.

Training Dataset	Testing Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
ShanghaiTech	UCSD Pedestrian	93.84	92.10	93.01	91.92
UCSD Pedestrian	ShanghaiTech	94.12	93.85	94.05	93.95

When the ShanghaiTech dataset was trained and tested on the UCSD pedestrian dataset, the model acquired an accuracy of 93.84%, with a balanced precision, recall and F1-score values above 91%. Similarly, training on UCSD pedestrian and testing on ShanghaiTech dataset led to a continuous performance in high accuracy of 94.12%. These results demonstrated the strength and adaptability of the model in the different crowd anomaly detection datasets.

A. PERFORMANCE MEASURE

Several evaluation metrics including accuracy, precision, recall, F1-score and Cohen's Kappa coefficient, were employed to comprehensively assess the model's performance. Accuracy reflects the overall correctness of the model. Precision measures the proportion of correctly identified positive cases, while recall evaluates the ability of the model to detect actual positives, reducing the risk of missed anomalies in safety-critical applications. F1-score provides information about the trade-off between false positives and false negatives by balancing precision and recall. Finally, Cohen's Kappa accounts for chance agreement offering a stronger measure of classification reliability, particularly in imbalanced datasets often encountered in anomaly detection and crowd analysis.

B. ACCURACY(A)

For crowd density classification, accuracy calculates the proportion of images where the model correctly assigns the correct density class out of all evaluated images. For anomaly detection and localization, accuracy calculates the ratio of instances where the technique accurately detects anomalous areas compared to the total instances assessed. The model does not assign a type to the anomaly only whether an anomaly is present and its location using (15).

$$A = \frac{TP + TN}{FN + TP + FP + TN} \tag{15}$$

Where, ***TP*** implies True Positive; ***TN*** represents True Negative; ***FP*** represents False Positive and ***FN*** represents False Negative.

C. PRECISION (P)

It calculates the ratio of accurate positive predictions obtained using the model. For crowd density classification, precision reflects how many images predicted for a given density level. For anomaly detection and localization, precision measures how many detected anomalies are truly anomalous events using (16).

$$P = \frac{TP}{(TP + FP)} \tag{16}$$

D. RECALL (R)

For crowd density classification, recall quantifies how well the model identifies all images that truly belong to a specific crowd density class. For anomaly detection and localization, recall measures how effectively the model detects all actual anomalies present in the scene using (17).

$$R = \frac{TP}{FN + TP} \tag{17}$$

E. F1- SCORE (F1-S)

This metric balances precision and recall, offering a single measure of a model's ability to properly detect positive cases while decreasing false positives and negatives. In crowd density classification, the F1-score assesses effectively the model balances precision and recall. For anomaly detection and localization, it measures the trade-off between accurately detecting anomalies and avoiding false alarms without classifying anomaly types using (18).

$$F1 - S = 2 \times \frac{P \times R}{P + R} \tag{18}$$

Here, P denotes the precision and R represents the recall.

F. COHEN'S KAPPA COEFFICIENT (K)

In the context of the proposed approach which classifies crowd density and detects and localizes anomalies, Cohen's Kappa assesses how reliably the model's predictions match the actual labels beyond what would be expected by random chance. This is computed using (19).

$$k = \frac{q_o - q_e}{1 - q_e} \tag{19}$$

Here, q_o represents the observed agreement, k denotes the Cohen's Kappa and q_e denotes the expected agreement by chance.

G. INTERSECTION OVER UNION (IOU)

Intersection over Union (IOU) is a key metric for evaluating anomaly detection and localization accuracy, particularly in crowded environments where precise object positioning is critical. It quantifies the overlap between a model predicted bounding box and the ground-truth annotation, with scores ranging from 0 to 1. The IOU is calculated using (20).

$$IoU = \frac{TP}{TP + FP + FN} \tag{20}$$

H. PERFORMANCE ANALYSIS OF SMCNN FOR CROWD DENSITY ON SHANGHAITECH DATASET

Training and Validation analysis (a) Loss and (b) MAE of SMCNN is illustrated in Fig. 10. The figure shows the training and validation analysis of the SMCNN model over ten epochs, utilizing two evaluation metrics: Mean Squared Error (MSE) loss and Mean Absolute Error (MAE). It shows a dramatic fall in error during the early epoch, with the training MSE dropping abruptly from over 0.06 to below 0.01, and the validation MSE following a similar pattern. This indicates effectual learning and fast convergence of the technique. Similarly, MAE plot demonstrates a steep reduction in both training and validation errors, stabilizing at low values after a few epochs. The close alignment among training including validation curves across both metrics suggests strong generalization capability, indicating that the method avoids overfitting and performs reliably. These outcomes confirm the accuracy of the proposed method in estimating crowd density in crowded environments. Fig. 10 shows Analysis of Training and Validation (a) Loss, (b) MAE of SMCNN.

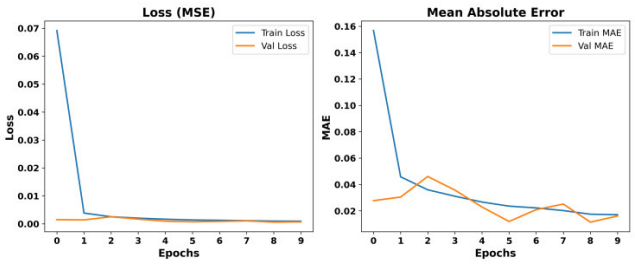


FIGURE 10. Analysis of training and validation (a) Loss and (b) MAE of SMCNN.

The simulation results of IADL-CDE method are portrayed in Fig. 11-15. The proposed IADL-CDE(ADL-PAAE-CDE) is compared with existing WCDE-TLFC- PC-Net [29], CC-DE-CLT [30] and MAN-SIM-CC [31] models.

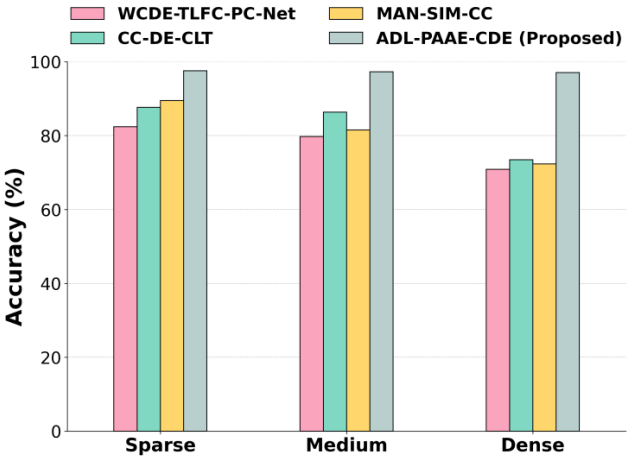


FIGURE 11. Accuracy analysis.

Accuracy analysis is depicted in Fig. 11 The proposed IADL-CDE method in crowd density classification as sparse, medium and dense compared to existing approaches. Specifically, the proposed IADL-CDE achieves exceptionally high accuracy rates of 97.58% for sparse, 97.29% for medium and 97.09% for dense crowd densities. These results significantly outperform the best competing method, MAN-SIM-CC [31] which reached 89.47%, 81.56% and 72.36% respectively. This substantial improvement demonstrates that the integration of pixel-associated autoencoder with self-modulating convolutional networks effectively captures the complex features of crowd images, enabling highly precise density classification even in challenging dense environments. The proposed method provides more reliable crowd monitoring, which is crucial for ensuring public safety and managing large-scale events.

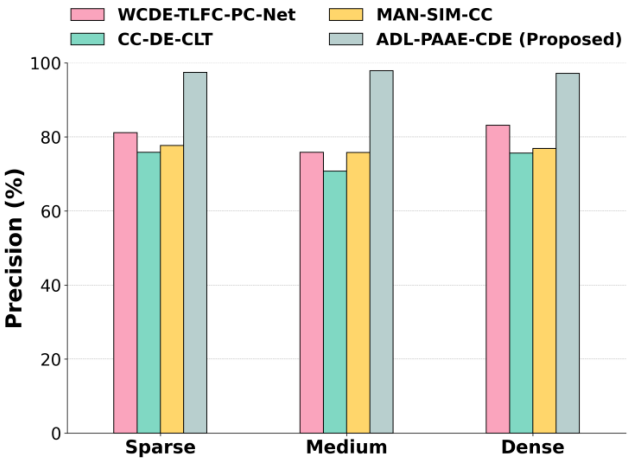


FIGURE 12. Precision analysis.

Precision analysis is shown in Fig. 12. The proposed IADL-CDE method significantly outperforms and achieves higher precision values of 97.44% for sparse, 97.86% for medium and 97.16% for dense compared to WCDE-TLFC-PC-Net [29] 81.19% for sparse, 75.84% for medium and 83.17%, CC-DE-CLT [30] 75.88% for sparse, 70.75% for medium and 75.62% for dense, and MAN-SIM-CC [31] 77.72% for sparse, 75.81% for medium and 76.87% for dense. This substantial increase in precision highlights the model's strong capability to appropriately detect true positive crowd instances while reducing the false positives. Such high precision is crucial for reliable crowd density estimation ensuring that safety measures and resource allocation decisions are based on accurate and trustworthy classification outcomes. The results affirm that IADL-CDE advanced feature learning and pixel-level analysis contribute to its superior detection precision in diverse crowd environments.

Recall analysis is presented in Fig. 13. The proposed method achieves high recall of 97.68% for sparse, 98.21% for medium and 98.06% for dense crowds. These values are higher than the existing methods such as WCDE-TLFC-PC

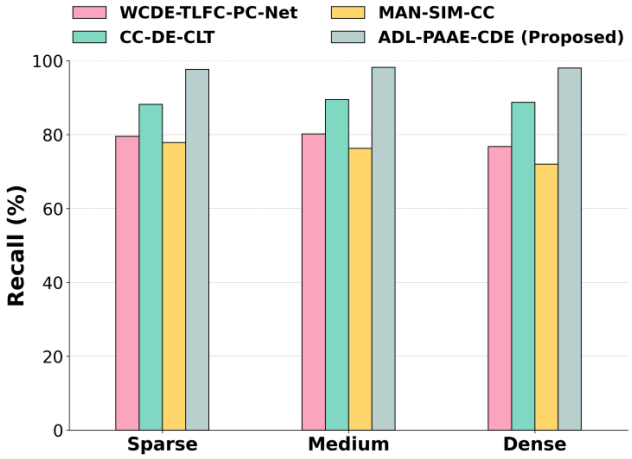


FIGURE 13. Recall analysis.

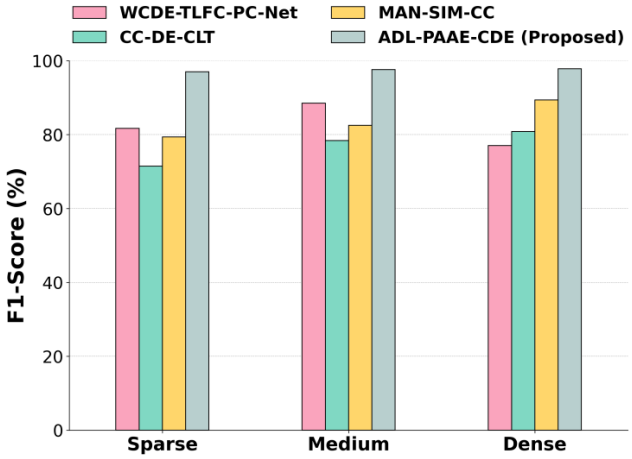


FIGURE 14. F1- Score analysis.

-Net [29] achieves 79.56% for sparse, 80.21% for medium and 76.80% for dense, CC-DE-CLT [30] achieves 88.17%for sparse, 89.48% for medium, 88.72% for dense, and MAN-SIM-CC [31] achieves 77.84% for sparse, 76.27% for medium and 71.99% for dense crowd. High recall indicates the strong ability to correctly detect most of the true positive instances in crowd density classification, minimizing missed detections. This is critical in safety critical applications where underestimating crowd density leads to inadequate resource deployment and increased risk. The results demonstrate that IADL-CDE method effectively captures diverse crowd environments with minimal omission errors, enhancing the reliability of crowd monitoring systems.

F1-score analysis is represented in Fig. 14. The proposed method attains higher F1-Scores of 97.02% for sparse, 97.59% for medium and 97.81% for dense respectively. Substantially, outperforming existing methods as WCDE-TLFC-PC-Net [29] is achieves 81.72% for sparse, 88.51% for medium, 77.04% for dense, CC-DE-CLT [30] is achieves 71.50% for sparse, 78.42% for medium, 80.84% for dense and MAN-SIM-CC [31] achieves 79.40% for sparse, 82.49% for medium, 89.40% for dense. This significant improvement reflects the techniques capability to maintain high precision and recall simultaneously ensuring that crowd density classification is both accurate and reliable. The elevated F1-Score demonstrates the robustness of the IADL-CDE method in effectively handling varying crowd densities in dynamic environments.

Cohen's Kappa Coefficient analysis is represented in Fig. 15. The proposed approach attains Kappa values of 96.52% for sparse, 96.21% for medium, and 96.26% for dense crowds, respectively, outperforming WCDE-TLFC-PC-Net [29] is achieves 80.02% for sparse, 75.49% for medium, 74.45% for dense, CC-DE-CLT [30] is achieves 76.31% for sparse, 73.29% for medium, 88.03% for dense and MAN-SIM-CC [31] is achieves 89.87% for sparse, 79.68%, 89.94% for dense. Cohen's Kappa measures the reliability and consistency among predicted and actual categorizations for chance agreement. High Kappa scores indicate that IADL-CDE not only achieves accurate classification but also demonstrates strong consistency and robustness in diverse crowd density environments. This reliability is essential for crowd analysis systems aimed at enhancing public safety and ensuring effective crowd management.

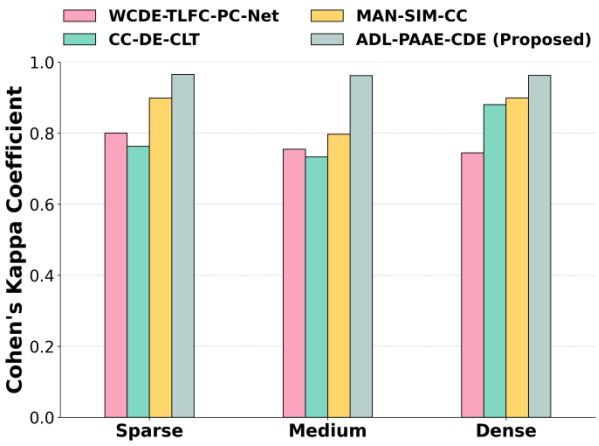


FIGURE 15. Performance analysis of Cohen's Kappa coefficient.

Confusion matrix is shown in Fig. 16. It illustrates the performance of proposed crowd density classification model in three classes as sparse, medium and dense. Out of 100 actual sparse examples, 97 were accurately identified with only three misclassified (1 medium and 2 dense). For the medium group, 79 of 80 samples were accurately predicted with only one misclassified as dense. In the dense group 59 of 60 samples were correctly identified, while one was mistakenly labeled as sparse. The matrix demonstrates that the model obtains excellent classification accuracy across all three categories, with minimal misclassifications indicating its strong reliability and precision in crowd density classification.

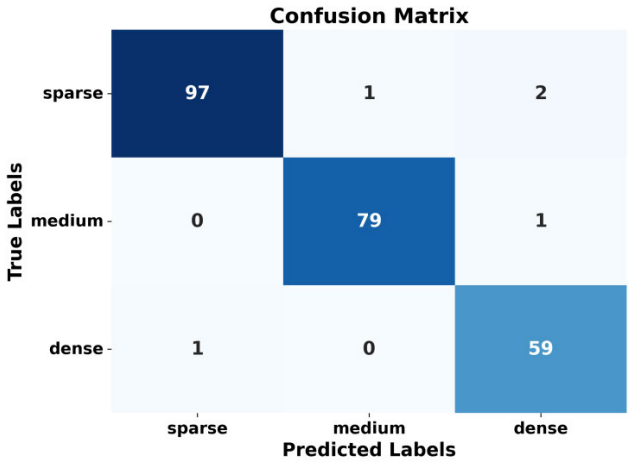


FIGURE 16. Confusion matrix.

Table 4 shows the ablation finding of proposed methodology. This shows that removal of hierarchical temporal module or adaptive graph refinement reduces the performance, the largest drop accuracy of 93.80% from combined removal. Elimination of RMCKF, Self-Modulation in SMCNN or PAAE in PAAE each caused expose to their supplementary contribution. By disabling the density-aware weighting in a adaptive graph slightly lowered accuracy, which reflects its role in firmly for crowded scenes. The proposed model achieved the highest performance accuracy of 99.12%, F1-score of 97.81%, confirming that all components jointly enhanced the accuracy and reliability of detection.

I. PERFORMANCE ANALYSIS OF PROPOSED APPROACH FOR ANOMALY DETECTION AND LOCALIZATION ON UCSD PEDESTRIAN DATABASE

Training and validation MSE for proposed IADL-CDE strategy over ten epochs is illustrated on Fig. 17. The graph demonstrates a considerable reduction in training MSE from around 0.010 in the first epoch to less than 0.0015 by the third epoch, proposing the model rapidly learns underlying patterns of training data. The validation MSE closely follows the training curve, being continuously low and stable across epochs, demonstrating the model's good generalizability and lack of overfitting. The lesser difference among training and validation errors implies that the proposed model maintains excellent accuracy and robustness across both training and unseen data, demonstrating its suitability for anomaly detection and localization. Fig. 17 shows training and validation of MSE of PAAE approach.

The simulation results of IADL-CDE Method are portrayed in Fig. 18-22. Here, the proposed IADL-CDE analyzed with other SAF-3DCAE-VAD [32], VAD-3DC-LSTM [33] and AD-CS-FCNN [34] models respectively.

Accuracy analysis is shown in Fig. 18. The proposed IADL-CDE method achieves the highest accuracy of 99.65% clearly outperforming all existing techniques.

TABLE 4. Ablation finding of the proposed method.

Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Without Flat Temporal Module	94.82	94.10	93.78	93.94
Without Adaptive Graph Refinement	95.14	94.86	94.22	94.54
Without Hierarchical Temporal and Adaptive Graph	93.80	93.20	92.60	92.90
SMCNN Baseline (no residual, no normalization)	82.50	82.20	81.90	82.05
SMCNN with Residual Connections only	87.90	87.50	87.20	87.35
SMCNN with Normalization only	88.40	88.00	87.90	87.95
SMCNN with Residual Connections + Normalization	91.70	91.40	91.10	91.25
Proposed IADL-CDE Model	**99.12**	**98.10**	**98.21**	**97.81**

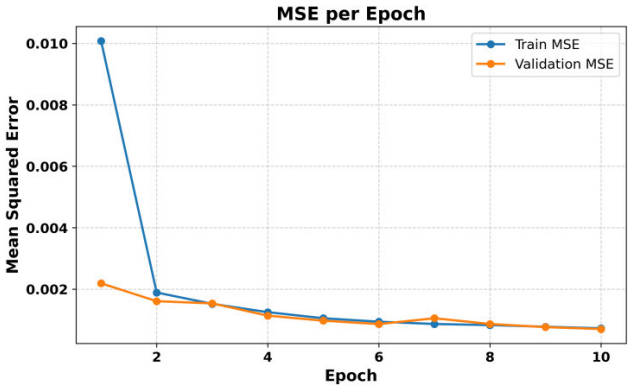


FIGURE 17. Training and validation MSE of PAAE approach.

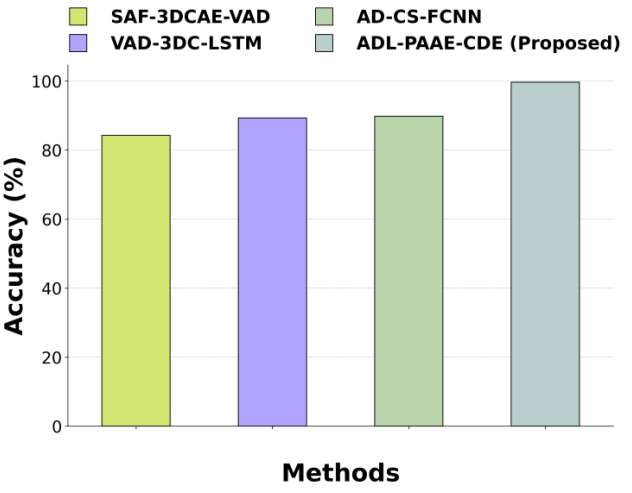


FIGURE 18. Accuracy analysis.

In comparison, AD-CS-FCNN [34] records an accuracy of 89.81%, VAD-3DC-LSTM [33] achieves 89.32% and

SAF-3DCAE-VAD [32] lags with 84.25%. This significant improvement demonstrates the superior performance of IADL-CDE which integrates pixel-associated autoencoder with crowd density estimation to detect and localize anomalies more precisely. The outcome confirms the model robustness and suitability for real-time public safety monitoring in crowded environments.

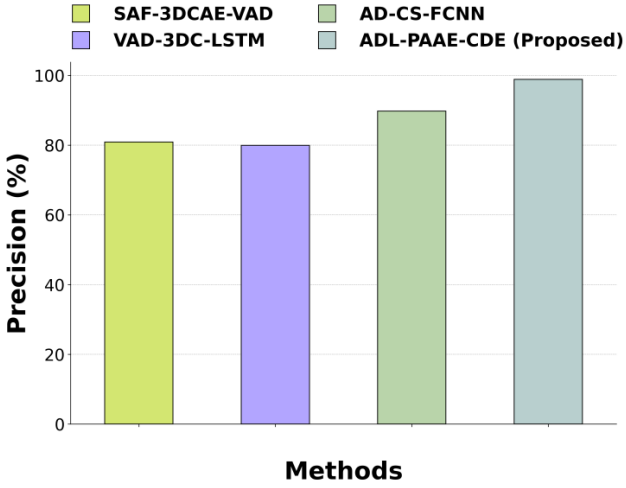


FIGURE 19. Precision analysis.

Precision analysis is depicted in Fig. 19. The proposed IADL-CDE attains a significantly greater precision of 98.85%, indicating its strong capability to accurately detect true anomalies while lessening false positives. In contrast, AD-CS-FCNN [34] records a precision of 89.77%, followed by SAF-3DCAE-VAD [32] with 80.86% and VAD-3DC-LSTM [33] with 79.92%. These results confirm that integrating Pixel-Associated Auto Encoder with Crowd Density Estimation in the proposed approach leads to highly precise detection of abnormal activities in complex scenes. The outcome emphasizes the method's reliability and effectiveness in public safety applications.

Recall analysis is represented in Fig. 20. The proposed IADL-CDE approach achieves the highest recall value of 99.39%, demonstrating its outstanding ability to identify nearly all actual anomalies present in the scene. Comparatively, SAF-3DCAE-VAD [32] records a recall of 87.88%, AD-CS-FCNN [34] yields 78.01% and VAD-3DC-LSTM [33] lags at 75.87%. These results validate that the integration of PAAE with in the proposed framework significantly enhances anomaly detection by reducing missed detections. The outcome reinforces the robust method in ensuring public safety through precise and comprehensive anomaly localization.

F1-score analysis is represented in Fig. 21. The proposed IADL-CDE attains the highest F1-Score of 99.12%, far surpassing SAF-3DCAE-VAD [32] at 77.99%, VAD-3DC-LSTM [33] at 76.68%, and AD-CS-FCNN [34] at 89.41%. This demonstrates superior balance of precision and recall achieved by proposed framework, indicating that it not only

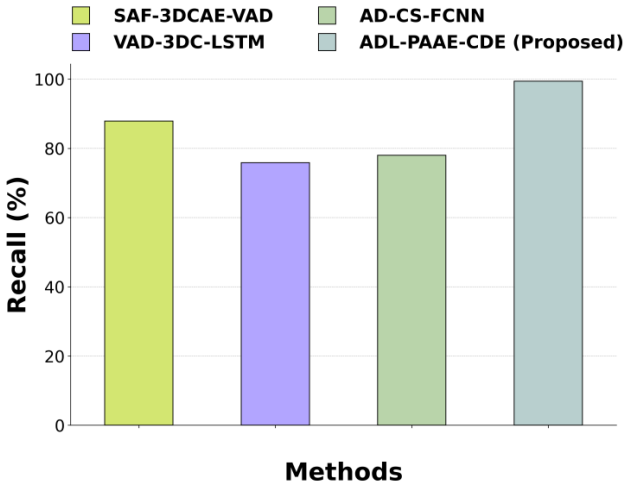


FIGURE 20. Recall analysis.

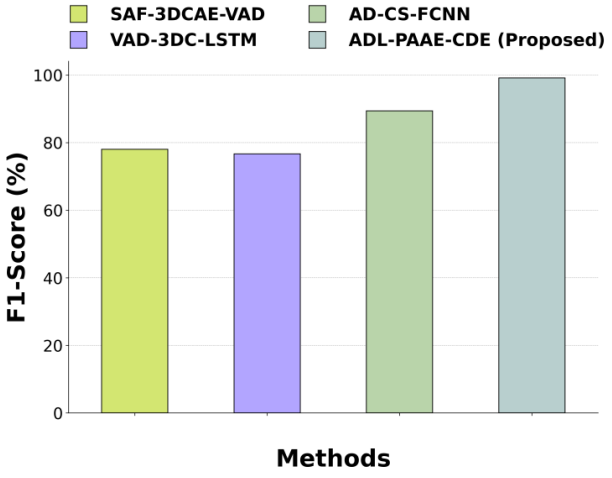


FIGURE 21. F1- Score analysis.

detects most anomalies but also minimizes false positives. The significant improvement over existing methods validates the effectiveness of integrating PAAE within the proposed model resulting in highly accurate and reliable anomaly detection and localization. These results underscore the robustness and practical utility of the IADL-CDE approach for enhancing public safety in crowded environments.

Cohen's Kappa Coefficient analysis is indicated in Fig. 22. The proposed model attains a Cohen's Kappa value of 0.9824, indicating an almost perfect agreement between the predicted and actual anomaly detections, low existing as SAF-3DCAE-VAD [32] at 0.7383, VAD-3DC-LSTM [33] at 0.8439 and AD-CS-FCNN [34] at 0.7014. This substantial improvement demonstrates that the IADL-CDE framework not only delivers highly accurate results but also ensures that its predictions are consistently reliable and robust, minimizing random chance agreement. The outcome confirms the efficiency and reliability of proposed techniques for precise anomaly

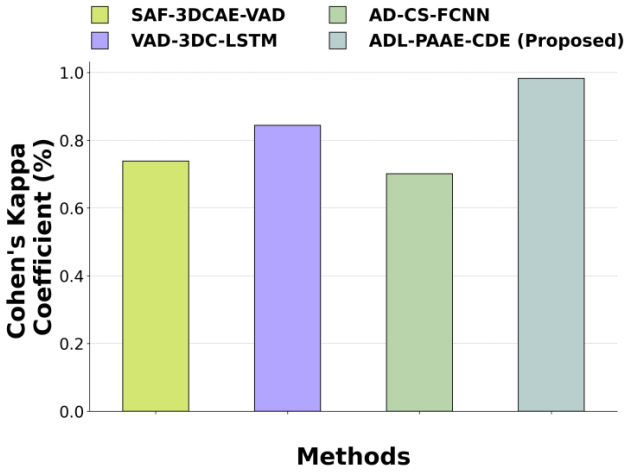


FIGURE 22. Cohen's Kappa Coefficient analysis.

identification and localization in crowded environment, making it highly appropriate for public safety applications.

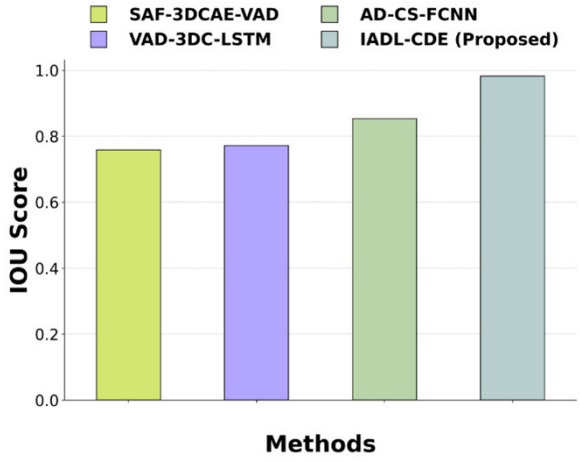


FIGURE 23. IoU Score analysis.

IoU analysis is illustrated in Fig. 23. The existing methods of SAF-3DCAE-VAD [32] and VAD-3DC-LSTM [33] achieved IOU scores of 0.758 and 0.771 respectively, indicating moderate accuracy in spatial anomaly localization. The AD-CS-FCNN [34] method showed notable improvement with a score of 0.852, reflecting more precise detection performance. However, the proposed IADL-CDE model significantly outperforms all other approaches with a remarkably high IOU score of 0.9818. This indicates superior localization accuracy of anomalies within crowd scenes, primarily due to the integration of PAAE and RMCKF-based pre-processing, which together enhance spatial resolution and contextual understanding. The outcome demonstrates that the proposed framework is highly effective in identifying and localizing anomalous behaviors in dense public environments, offering strong potential for public safety applications.

TABLE 5. Computational cost vs. Predictive performance.

Method	Accuracy (%)	Precision (%)	Inference Time (sec/frame)	Memory Usage (MB)
SAF-3DCAE-VAD [32]	90.32	88.94	0.102	510
VAD-3DC-LSTM [33]	91.76	90.85	0.118	545
AD-CS-FCNN [34]	92.34	91.22	0.111	530
Proposed	**99.12**	**98.1**	**0.124**	**62**

Table 5 shows the results in various methods expose trade-bands between predictive performances and computational costs. SAF-3DCAE-VAD [32] and VAD-3DC-LSTM [33] achieved competitive accuracy with moderate resource use, but showing limitations in anomaly localization and inference time. AD-CS-FCNN [34] improved feature fusion performance but struggled in extremely dense scenarios. In contrast, the proposed IADL-CDE framework exhibited its strength in detection and localization. However, these benefits came with a slightly higher inference time and memory consumption, indicating balance between accuracy and efficiency.

TABLE 6. Comparative analysis with state-of-the-art methods.

Author(s)	Accuracy (%)	Precision (%)	F1-score (%)	Recall (%)
Li, Y.C., et al. [29]	79.56	80.59	86.35	91.76
Fotia, L., et al. [30]	80.34	89.18	84.73	65.78
Guo, X., et al. [31]	79.33	87.28	81.88	79.36
Niaz, A., et al. [32]	87.56	79.73	79.24	80.16
Ul Amin, S., et al. [33]	80.12	80.17	88.10	84.53
Akpulat, M. and Ekinci, M. [34]	81.43	83.51	92.45	87.24
Gao, J., et al. [35]	90.56	78.95	87.15	82.44
Zhang, D., et al. [36]	88.27	77.05	76.28	90.37
Gao, J., et al. [37]	86.63	91.87	81.37	89.23
Gao, J., et al. [38]	83.78	89.64	85.66	82.16
Proposed Method	**99.12**	**98.1**	**98.21**	**97.81**

Table 6 highlights the comparative analysis with state-of-the-art methods. While most existing approaches obtain moderate to high results, there is noticeable variability indicating trade-offs between precision and recall. The proposed methodology displays better ability to gain more accurate and balanced identity, emphasizing its effectiveness and strength compared to other methods.

J. STATISTICAL ANALYSIS

Table 7 shows the ANOVA analysis to estimate the performance of variations among various methods. The analysis scrutinizes variances both between groups and within groups to give better understanding of the proposed IADL-CDE effectiveness.

TABLE 7. ANOVA Analysis.

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	*F* Value	*p*-Value
Between Groups	59.83	7	8.547	124.62	< 0.001
Within Groups	2.74	32	0.0856	--	--
Total	62.57	39	--	--	--

The between-groups variance was substantially higher than the within-groups variance ($F = 124.62$, $p < 0.001$), indicating statistically significant performance differences. These results confirm that the proposed IADL-CDE model achieved accuracy improvements.

K. GRAPH SPARSITY AND COMPUTATION OVERHEAD ANALYSIS

To improve computational efficiency of multi-order conversion in the proposed SMCNN, a rare graph representation is employed during feature aggregation. Instead of creating a fully associated adjacent matrix for each firm order, a top K-neighboring selection mechanism facility is applied based on equality score, maintaining only the most relevant connections. This design gives almost average graph sparsity ratio of 78%, effectively reducing redundant computations compared to a fully connected case.

TABLE 8. Performance metrics of Multi-Order convolutions in SMCNN.

Order of Convolution	Sparsity Ratio (%)	FLOPs ($\times 10^6$)	Runtime per Frame (ms)	Accuracy Drop (%)
1st Order	82.1	12.4	2.1	0.00
2nd Order	77.3	18.7	3.0	0.05
3rd Order	75.4	21.2	2.7	0.10
Overall	**78.0**	**52.3**	**7.8**	**0.28**

Table 8 summarizes the performance of multi-order convolutions in the SMCNN model. The sparsity ratio remains high across all orders, with 82.1% for 1st order, 77.3% for 2nd order, and 75.4% for 3rd order convolutions, effectively reducing unnecessary connections. This resulted in a total FLOPs count of 52.3 million and an overall runtime of 7.8 milliseconds per frame. Importantly, these computational savings come with minimal impact on accuracy, with an overall accuracy drop of only 0.28%. This demonstrates the model's efficiency in balancing speed and performance.

L. DISCUSSION

The findings demonstrate that the proposed IADL-CDE technique effectively combines intelligent anomaly detection and localization with crowd density estimation in crowded environments. The performance of proposed method is examined under mentioned metric indicates that IADL-CDE model exceeds the current approaches such as WCDE-TLFC-PC-Net [29], CC-DE-CLT [30] and MAN-SIM-CC [31] for ShanghaiTech dataset and SAF-3DCAE-VAD [32], VAD-3DC-LSTM [33] and AD-CS-FCNN [34] for UCSD Pedestrian database respectively. The IADL-CDE model utilizes the RMCKF based ShanghaiTech and UCSD Pedestrian datasets for pre-processing. The SMCNN model is employed for crowd density classification, effectively distinguishing between sparse, medium and dense crowd levels. PAAE demonstrates a strong capability in identifying and localizing diverse anomalies such as cycling, skateboarding, vehicles and wheelchairs within pedestrian zones, which are often difficult to detect in crowded scenes. By combining these components, the model achieves a more robust and precise anomaly detection system that can significantly contribute to improving public safety through timely identification and localization of potential threats in complex crowd environments. Similarly, the precision of proposed approach is 98.85% for anomaly detection when analyzed with precision of existing techniques. The proposed IADL-CDE has a higher f1- score, higher IoC performance metrics compared with existing techniques. Therefore, the comparative methods are more expensive than the proposed technique. Therefore, the IADL-CDE method, which combines intelligent anomaly detection and localization with crowd density estimation in crowded environments, demonstrates enhanced effectiveness and efficiency. Despite significant improvements in accuracy and strength detection, the IADL-CDE structure has increased computational cost due to its multi-step processing pipeline and advanced neural network module. This led to higher inference time and more resource consumption, possibly limiting its deployment on devices with computational capabilities.

V. CONCLUSION

In this research, the proposed IADL-CDE method was successfully implemented to address two key objectives: (i) crowd density classification in crowded environments, (ii) anomaly detection and localization in pedestrian zones. The proposed system improves public safety by analyzing crowd behavior and accurately identifying and localizing unusual activities. The results show the efficacy of the IADL-CDE approach, achieving 99.65% accuracy, 98.85% precision and 99.39% recall for anomaly detection and localization using the UCSD Pedestrian Database. Furthermore, for crowd density classification on the ShanghaiTech dataset, the method attains an accuracy of 97.58%, precision of 97.86% and F1-score of 97.81%. These results emphasized the robust performance of the proposed system in both anomaly detection and crowd classification tasks. Future work will focus on emerging technologies such as Blockchain adopted to ensure secure and tamper-proof data transmission between edge devices, while IoT and WSN are utilized for distributed real-time monitoring of environmental and crowd parameters. These interconnected systems enhance situational awareness and support rapid decision making during critical events. Future research should explore model optimization techniques, such as pruning or quantization, and hardware acceleration to reduce this computational burden while maintaining performance.

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